

JACOB LUSTIG-YAEGER

Exoplanet Astronomer | Johns Hopkins APL | Astronomy & Astrobiology, PhD

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Education PhD in Astronomy & Astrobiology (dual-titled PhD) (2014 – 2020)
MS in Astronomy (2016)
University of Washington, Seattle, WA
Thesis: “[The Detection, Characterization, and Retrieval of Terrestrial Exoplanet Atmospheres](#)”
Advisor: Professor Victoria Meadows

BS in Physics (Honors), Minor in Mathematics (2009 – 2013)
University of California, Santa Cruz, CA
Advisor: Professor Jonathan Fortney

Research Interests — Characterizing terrestrial exoplanets for habitability and signs of life
— Retrieving terrestrial planet atmospheres using Bayesian inference
— Modeling telescope sensitivity to motivate future exoplanet science cases
— Developing novel methods to detect exoplanet habitability and biosignatures

Research Experience *Senior Professional Staff:* The Johns Hopkins Applied Physics Laboratory (2023 – present)
Exoplanet Astronomer, JWST GO & NASA program PI

Post Doctoral Fellow: The Johns Hopkins Applied Physics Laboratory (2020 – 2023)
Terrestrial exoplanet atmospheres, theory & observation, mission concepts

Graduate Research Assistant: Virtual Planetary Laboratory (2014 – 2020)
Terrestrial exoplanets, their atmospheres, habitability & biosignatures

Undergraduate & Postbaccalaureate Researcher: University of California, Santa Cruz (2012 – 2014)
Hot Jupiter & brown dwarf atmospheres, opacity sources, and atmospheric retrieval

Teaching Experience *Research Mentor:* Department of Astronomy, University of Washington (Sept 2016 – 2020)
Advising undergraduate students in exoplanet astronomy and astrobiology research

Teaching Assistant: Department of Astronomy, University of Washington (Sept 2014 – June 2015)
Led two biweekly sections for undergraduate students
— ASTR 101 (Spring 2015; Autumn 2014); ASTR 150 (Winter 2015)

Math & Writing Tutor: Learning Support Services, UCSC (Sept 2010 – June 2012)
Instructed students in college level mathematics and writing as a group and drop-in tutor

Honors, Awards, & Funded Proposals — **JWST Cycle 4 PI:** “[Charting the Cosmic Shoreline](#)” (2025)
— **NASA Group Achievement Award:** Parker Solar Probe Team (2023)
— **NASA Exoplanet Research Program PI:** “[Venus as an Exoplanet: Constructing a Venus Transmission Spectrum from Venus Express Solar Occultation Measurements](#)” (2022)
— **JWST Cycle 1 Co-PI:** “[Tell Me How I’m Supposed To Breathe With No Air: Measuring the Prevalence and Diversity of M-Dwarf Planet Atmospheres](#)” (2021)
— **NASA Group Achievement Award:** LUVIR Mission Concept Study Team (2019)
— Honors undergraduate thesis in physics (2013)
— University Honor, *cum laude* at University of California, Santa Cruz (2013)

Technical Skills

- Programming in Python, Julia, IDL, & Fortran
- Packaging Python code for open-source distribution
- Bayesian parameter inference and model selection
- Version control and collaboration with Git & GitHub
- Data visualization with Matplotlib, Tableau, & Bokeh
- Writing and typesetting with $\text{\LaTeX}$ and Overleaf
- Extensive experience with MS Office Suite

Community Service

- HWO steering committee member for Living Worlds, Characterizing Exoplanets, and Exoplanet Science Yield working groups (Feb 2024 - present)
- Member of the Exoplanet Science Metrics Working Group (May 2023 - April 2024)
- Science Organizing Committee: [STScI 2023 Spring Symposium](#) (Aug 2022 – 2023)
- Section Co-Editor: “Handbook of Exoplanets, Second Edition” (Mar 2022 – present)
- Participating Member and Task Co-Lead: [ExoPAG SAG 22](#) (Oct 2020 – Dec 2021)
- Reviewer for AAS Journals, MNRAS, Nature Astronomy (2017 - present)

Publications *First Authored*

11. **Lustig-Yaeger, J.**, Cowan, N. B., Hu, R., Mayorga, L. C., & Robinson, T. D. (2026). “[Detecting Surface Liquid Water on Exoplanets](#)”. *Astronomical Society of the Pacific Conference Series*, 542, 439.
10. **Lustig-Yaeger, J.**, Sotzen, K. S., Stevenson, K. B., Tsai, S.-M., Challener, R. C., et al. (2025). “[JWST-TST DREAMS: The Nightside Emission and Chemistry of WASP-17b](#)”. *The Astrophysical Journal Letters*, 994, L4.
9. **Lustig-Yaeger, J.**, Izenberg, N. R., Gilmore, M. S., Mayorga, L. C., May, E. M., Vourlidas, A., Hess, P., Wood, B. E., Howard, R. A., Raouafi, N. E., Arney, G. N. (2023). “[A WISPR of the Venus Surface: Analysis of the Venus Nightside Thermal Emission at Optical Wavelengths](#)”. *The Planetary Science Journal*, 4(11), 207.
8. **Lustig-Yaeger, J.**, Meadows, V. S., Crisp, D., Line, M. R., & Robinson, T. D. (2023). “[Earth as a Transiting Exoplanet: A Validation of Transmission Spectroscopy and Atmospheric Retrieval Methodologies for Terrestrial Exoplanets](#)”. *The Planetary Science Journal*, 4(9), 170.
7. **Lustig-Yaeger, J.** & Fu, G., May, E. M., Ceballos, K. N., Moran, S. E., Peacock, S., Stevenson, K. B., Kirk, J., López-Morales, M., MacDonald, R. J., Mayorga, L. C., Sing, D. K., et al. (2023). “[A JWST transmission spectrum of the nearby Earth-sized exoplanet LHS 475 b](#)”. *Nature Astronomy*, 1-12. <https://doi.org/10.1038/s41550-023-02064-z>
6. **Lustig-Yaeger, J.**, Sotzen, K. S., Stevenson, K. B., Luger, R., May, E. M., Mayorga, L. C., Mandt, K., & Izenberg, N. R. (2022). “[Hierarchical Bayesian Atmospheric Retrieval Modeling for Population Studies of Exoplanet Atmospheres: A Case Study on the Habitable Zone](#)”. *The Astronomical Journal*, 163(3), 140.
5. **Lustig-Yaeger, J.**, Stevenson, K. B., Mayorga, L. C., Sotzen, K. S., May, E. M., Izenberg, N. R., & Mandt, K. (2021). “[Retrieving Exoplanet Atmospheres using Planetary Infrared Excess: Prospects for the Night side of WASP-43 b and other Hot Jupiters](#)”. *The Astrophysical Journal Letters*, 921(1), L4.
4. **Lustig-Yaeger, J.**, Meadows, V. S., & Lincowski, A. P. (2019). “[A Mirage of the Cosmic Shoreline: Venus-like Clouds as a Statistical False Positive for Exoplanet Atmospheric Erosion](#)”. *The Astrophysical Journal Letters*, 887(1), L11.
3. **Lustig-Yaeger, J.**, Robinson, T. D., & Arney, G. (2019). “[coronagraph: Telescope Noise Modeling for Exoplanets in Python](#)”. *Journal of Open Source Software*, 4(40), 1387.
2. **Lustig-Yaeger, J.**, Meadows, V. S., & Lincowski, A. P. (2019). “[The Detectability and Characterization of the TRAPPIST-1 Exoplanet Atmospheres with JWST](#)”. *The Astronomical Journal*, 158(1), 27.
1. **Lustig-Yaeger, J.**, Meadows, V. S., Tovar Mendoza, G., Schwieterman, E. W., Fujii, Y., Luger, R., & Robinson, T. D. (2018). “[Detecting Ocean Glint on Exoplanets Using Multiphase Mapping](#)”. *The Astronomical Journal*, 156(6), 301.

49. Mukherjee, S., Sing, D. K., Fu, G., Stevenson, K. B., Schmidt, S. P., et al. (2026). “Cloudy mornings and clear evenings on a gas giant exoplanet”. *Science*, 392, 858.
48. Gillon, M., Ducrot, E., Bell, T. J., Huang, Z., Lincowski, A. P., et al. (2026). “No thick atmosphere around TRAPPIST-1 b and c from JWST thermal phase curves”. *Nature Astronomy*, 10, 674.
47. Ashtari, R., **Lustig-Yaeger, J.**, Libby-Roberts, J., Mller, S., Kanodia, S., et al. (2026). “GEMS JWST: HATS-75 b A Giant Planet with a Subsolar Metallicity Atmosphere Orbiting an M Dwarf”. *The Astronomical Journal*, 171, 294.
46. Cañas, C. I., **Lustig-Yaeger, J.**, Tsai, S.-M., Mller, S., Helled, R., et al. (2026). “GEMS JWST: Transmission Spectroscopy of TOI-5205b Reveals Significant Stellar Contamination and a Metal-poor Atmosphere”. *The Astronomical Journal*, 171, 260.
45. Limbach, M. A., Dacus, B., Kotten, B., Lane, E., **Lustig-Yaeger, J.**, et al. (2026). “Exomoons and Exorings with the Habitable Worlds Observatory. II. Finding ”Endor” with Lunar Eclipses”. *The Astronomical Journal*, 171, 201.
44. Sotzen, K. S., **Lustig-Yaeger, J.**, Stevenson, K. B., Tsai, S.-M., Challener, R. C., et al. (2025). “Non-detection of Nightside Emission from HAT-P-26b due to Stellar Contamination”. *Research Notes of the American Astronomical Society*, 9, 352.
43. Stevenson, K. B., **Lustig-Yaeger, J.**, May, E. M., Kopparapu, R. K., Fauchez, T. J., et al. (2025). “K2-18b Does Not Meet the Standards of Evidence for Life”. *The Astronomical Journal*, 170, 257.
42. Bennett, K. A., MacDonald, R. J., Peacock, S., Perez, J., May, E. M., Moran, S. E., Alderson, L., **Lustig-Yaeger, J.**, Wakeford, H. R., Sing, D. K., et al. (2025). “Additional JWST/NIRSpec Transits of the Rocky M Dwarf Exoplanet GJ 1132 b Reveal a Featureless Spectrum”. *The Astronomical Journal*, 170, 205.
41. D’Incecco, P., Filiberto, J., Garvin, J. B., Arney, G. N., Getty, S. A., et al. (2025). “The “Analogues for VENUS’ GEologically Recent Surfaces” (AVENGERS) Initiative: Using Terrestrial Analogs to Study Recent Volcano-Tectonic Activity on Venus and Inform the Characterization of Terrestrial Exoplanets”. *Journal of Geophysical Research (Planets)*, 130, e2024JE008675.
40. DiTomasso, V., et al. (2025). “The Magnetically Induced Radial Velocity Variation of Gliese 341 and an Upper Limit to the Mass of Its Transiting Earth-sized Planet”. *The Astrophysical Journal*, 979, 214.
39. Mandt, K. E., **Lustig-Yaeger, J.**, Luspai-Kuti, A., Wurz, P., Bodewits, D., Fuselier, S. A., Mousis, O., Petrinec, S. M. and Trattner, K. J., (2024). “A nearly terrestrial D/H for comet 67P/Churyumov-Gerasimenko”. *Science Advances*, 10(46), p.eadp2191.
38. Deming, D., Fu, G., Bouwman, J., Dicken, D., Espinoza, N., Glasse, A., Greene, T., Kendrew, S., Law, D., **Lustig-Yaeger, J.** & Marin, M. G. (2024). “Toward Exoplanet Transit Spectroscopy Using JWST/MIRI’s Medium Resolution Spectrometer”. *Publications of the Astronomical Society of the Pacific*, 136(8), p.084402.
37. TRAPPIST-1 JWST Community Initiative, et al. (2024). “A roadmap for the atmospheric characterization of terrestrial exoplanets with JWST”. *Nature Astronomy*, 8, 810-818.
36. Carter, A. & May, E. M., et al. (2024). “A benchmark JWST near-infrared spectrum for the exoplanet WASP-39 b”. *Nature Astronomy*, 8, 1008-1019.
35. Limbach, M. A., **Lustig-Yaeger, J.**, Vanderburg, A., Vos, J. M., Heller, R., Robinson, T. D. (2024). “Exomoons and Exorings with the Habitable Worlds Observatory. I. On the Detection of EarthMoon Analog Shadows and Eclipses”. *The Astronomical Journal*, 168, 57.
34. Gilbert-Janizek, S., Meadows, V. S., **Lustig-Yaeger, J.** (2024). “Retrieved Atmospheres and Inferred Surface Properties for Terrestrial Exoplanets Using Transmission and Reflected-light Spectroscopy”. *The Planetary Science Journal*, 5, 148.
33. Kirk, J., Stevenson, K., B., Fu, G., **Lustig-Yaeger, J.**, et al. (2024). “JWST/NIRCam Transmission Spectroscopy of the Nearby Sub-Earth GJ 341b”. *The Astronomical Journal*, 167, 90.

32. May, E. M., MacDonald, R. J., Bennett, K. A., Moran, S. E., Wakeford, H. R., Peacock, S., **Lustig-Yaeger, J.**, et al. (2023). “Double Trouble: Two Transits of the Super-Earth GJ 1132 b Observed with JWST NIRSpec G395H”. *The Astrophysical Journal Letters*, 959(1), L9.
31. Mayorga, L. C., **Lustig-Yaeger, J.**, & Stevenson, K. B. (2023). “On the Effectiveness of the Planetary Infrared Excess (PIE) Technique to Retrieve the Parameters of Multiplanet Systems around M Dwarfs: A Case Study on the TRAPPIST-1 System”. *The Astrophysical Journal*, 956(2), 74.
30. Meadows, V. S., Lincowski, A. P., & **Lustig-Yaeger, J.** (2023). “The Feasibility of Detecting Biosignatures in the TRAPPIST-1 Planetary System with JWST”. *The Planetary Science Journal*, 4(10), 192.
29. Chouqar, J., **Lustig-Yaeger, J.**, Benkhaldoun, Z., Szentgyorgyi, A., Jabiri, A., & Soubkiou, A. (2023). “Surface pressure impact on nitrogen-dominated USP super-Earth atmospheres”. *Monthly Notices of the Royal Astronomical Society*, 522(1), 648-659.
28. Schlawin, E., Challener, R., Mansfield, M., Rauscher, E., Adams, A., & **Lustig-Yaeger, J.** (2023). “Planet Eclipse Mapping with Long-term Baseline Drifts”. *The Astronomical Journal*, 165(5), 210.
27. Moran, S. E., Stevenson, K. B., Sing, D. K., MacDonald, R. J., Kirk, J., **Lustig-Yaeger, J.**, Peacock, S., Mayorga, L. C., Bennett, K. A., López-Morales, M., May, E. M., et al. (2023). “High Tide or Riptide on the Cosmic Shoreline? A Water-Rich Atmosphere or Stellar Contamination for the Warm Super-Earth GJ 486b from JWST Observations”. *The Astrophysical Journal Letters*, 948(1), p.L11.
26. Ahrer, E. M., Stevenson, K. B., Mansfield, M. et al. (2023). “Early Release Science of the exoplanet WASP-39b with JWST NIRCам”. *Nature* 614, 653-658.
25. Rustamkulov, Z., Sing, D. K., Mukherjee, S. et al. (2023). “Early Release Science of the exoplanet WASP-39b with JWST NIRSpec PRISM”. *Nature* 614, 659-663.
24. JWST Transiting Exoplanet Community Early Release Science Team (2023). “Identification of carbon dioxide in an exoplanet atmosphere”. *Nature* 614, 649-652.
23. Limbach, M. A., Vanderburg, A., Stevenson, K. B., Blouin, S., Morley, C., **Lustig-Yaeger, J.**, Soares-Furtado, M. and Janson, M., (2022). “A new method for finding nearby white dwarfs exoplanets and detecting biosignatures”. *Monthly Notices of the Royal Astronomical Society*, 517(2), pp.2622-2638.
22. Mandell, A. M., **Lustig-Yaeger, J.**, Stevenson, K., & Staguhn, J. (2022). “MIRECLE: Science Yield for a Mid-IR Explorer-Class Mission to Study Non-Transiting Rocky Planets Orbiting the Nearest M-Stars Using Planetary Infrared Excess”. *The Astronomical Journal*, 164, 176.
21. Mandt, K., Luspay-Kuti, A., **Lustig-Yaeger, J.**, Felton, R., & Domagal-Goldman, S. (2022). “TRAPPIST-1h as an Exo-Titan. I. The Role of Assumptions about Atmospheric Parameters in Understanding an Exoplanet Atmosphere”. *The Astrophysical Journal*, 930(1), 73.
20. Wood, B. E., Hess, P., **Lustig-Yaeger, J.**, Gallagher, B., Korwan, D., Rich, N., et al. (2022). “Parker Solar Probe imaging of the night side of Venus”. *Geophysical Research Letters*, 49(3), e2021GL096302.
19. Hinkel, N. R., Pepper, J., Stark, C. C., Burt, J. A., Ciardi, D. R., Hardegree-Ullman, K. K., **Lustig-Yaeger, J.**, Kopparapu, R., Mishra, L., Molaverdikhani, K., Pascucci, I., et al. (2021). “Final Report for SAG 22: A Target Star Archive for Exoplanet Science”. *arXiv preprint arXiv:2112.04517*.
18. Komacek, T. D., Kang, W., **Lustig-Yaeger, J.**, & Olson, S. L. (2021). “Constraining the Climates of Rocky Exoplanets”. *Elements*, 17, 4.
17. Mayorga, L. C., **Lustig-Yaeger, J.**, May, E. M., Sotzen, K. S., Gonzalez-Quiles, J., et al. (2021) “Transmission Spectroscopy of the Earth-Sun System to Inform the Search for Extra-solar Life”. *The Planetary Science Journal*, 2(4), 140.
16. Kopparapu, R., Arney, G., Haqq-Misra, J., **Lustig-Yaeger, J.**, & Villanueva, G. (2021). “Nitrogen dioxide pollution as a signature of extraterrestrial technology”. *The Astrophysical Journal*, 908(2), 164.

15. Mansfield, M., Schlawin, E., **Lustig-Yaeger, J.**, et al. (2020). “Eigenspectra: A Framework for Identifying Spectra from 3D Eclipse Mapping”. *Monthly Notices of the Royal Astronomical Society*.
14. Leung, M., Meadows, V. S., & **Lustig-Yaeger, J.** (2020). “High-Resolution Spectral Discriminants of Ocean Loss for M Dwarf Terrestrial Exoplanets”. *The Astronomical Journal*, 160(1), 11.
13. Chouqar, J., Zouhair, B., Jabiri, A., **Lustig-Yaeger, J.**, Soubkiou, A., & Szentgyorgyi, A. (2020). “Properties of Sub-Neptune Atmospheres: TOI-270 System”. *Monthly Notices of the Royal Astronomical Society*, 495(1), 962-970.
12. Guzewich, S. D., **Lustig-Yaeger, J.**, Davis, C. E., Kopparapu, R. K., Way, M. J., & Meadows, V. S. (2020). “The Impact of Planetary Rotation Rate on the Reflectance and Thermal Emission Spectrum of Terrestrial Exoplanets Around Sun-like Stars”. *The Astrophysical Journal*, 893, 140.
11. Lincowski, A. P., **Lustig-Yaeger, J.**, & Meadows, V. S. (2019). “Observing Isotopologue Bands in Terrestrial Exoplanet Atmospheres with the James Webb Space Telescope—Implications for Identifying Past Atmospheric and Ocean Loss”. *The Astronomical Journal*, 158(1), 26.
10. Luger, R., Agol, E., Foreman-Mackey, D., Fleming, D. P., **Lustig-Yaeger, J.**, & Deitrick, R. (2019). “STARRY: Analytic Occultation Light Curves”. *The Astronomical Journal*, 157(2), 64.
9. Lincowski, A. P., Meadows, V. S., Crisp, D., Robinson, T. D., Luger, R., **Lustig-Yaeger, J.**, & Arney, G.N. (2018). “Evolved Climates and Observational Discriminants for the TRAPPIST-1 Planetary System”. *The Astrophysical Journal*, 867(1), 76.
8. Meadows, V. S., Reinhard, C. T., Arney, G. N., Parenteau, M. N., Schwieterman, E. W., Domagal-Goldman, S. D., Lincowski, A. P., Stapelfeldt, K. R., Rauer, H., DasSarma, S., Hegde, S., Narita, N., Deitrick, R., **Lustig-Yaeger, J.**, Lyons, T. W., & Siegler, N. (2018). “Exoplanet Biosignatures: Understanding Oxygen as a Biosignature in the Context of Its Environment”. *Astrobiology*, 18, 630-662.
7. Meadows, V. S., Arney, G. N., Schwieterman, E. W., **Lustig-Yaeger, J.**, Lincowski, A. P., Robinson, T., Domagal-Goldman, S. D., Barnes, R. K., Fleming, D. P., Deitrick, R., Luger, R., Driscoll, P. E., Quinn, T. R., Crisp, D. (2018). “The Habitability of Proxima Centauri b: Environmental States and Observational Discriminants”. *Astrobiology*, 18, 133-189.
6. Luger, R., **Lustig-Yaeger, J.**, & Agol, E. (2017). “Planet-Planet Occultations in TRAPPIST-1 and Other Exoplanet Systems”. *The Astrophysical Journal*, 851(2), 94.
5. Fujii, Y., **Lustig-Yaeger, J.**, & Cowan, N. B. (2017). “Rotational Spectral Unmixing of Exoplanets: Degeneracies between Surface Colors and Geography”. *The Astronomical Journal*, 154(5), 189.
4. Luger, R., **Lustig-Yaeger, J.**, Fleming, D. P., Tilley, M. A., Agol, E., Meadows, V. S., Deitrick, R., & Barnes, R. (2017). “The Pale Green Dot: A Method to Characterize Proxima Centauri b using Exo-Aurorae”. *The Astrophysical Journal*, 837, 63.
3. Barnes, R., Deitrick, R., Luger, R., Driscoll, P. E., Quinn, T. R., Fleming, D. P., Arney, G., Crisp, D., Domagal-Goldman, S. D., Lincowski, A. P., **Lustig-Yaeger, J.**, & Schwieterman, E. (2017). “The Habitability of Proxima Centauri b I: Evolutionary Scenarios”. *arXiv preprint arXiv:1608.06919*.
2. Greene, T. P., Line, M. R., Montero, C., Fortney, J. J., **Lustig-Yaeger, J.**, & Luther, K. (2016). “Characterizing transiting exoplanet atmospheres with JWST”. *The Astrophysical Journal*, 817(1), 17.
1. Freedman, R. S., **Lustig-Yaeger, J.**, Fortney, J. J., Lupu, R. E., Marley, M. S., & Lodders, K. (2014). “Gaseous Mean Opacities for Giant Planet and Ultracool Dwarf Atmospheres over a Range of Metallicities and Temperatures”. *The Astrophysical Journal Supplement Series*, 214(2), 25.

Presentations *Invited Talks*

13. UW Earth and Space Sciences department colloquium, 12 Feb 2026
12. Sagan Summer Workshop (Caltech), 25 July 2025
11. UW Astrobiology Colloquium, 29 April 2025
10. Sagan Summer Workshop (Caltech), July 2024
9. ISSI review talk and workshop participant on “The Geoscience of Exoplanets: Going Beyond Habitability”, Bern, Switzerland, 8-12 April 2024
8. UC Riverside Astrobiology Seminar, 2 February 2024
7. Caltech Planetary Science Seminar, 30 January 2024
6. **Lustig-Yaeger, J.** (2023). “Advancing the Study of Rocky Planets by Leveraging Connections Between Exoplanet and Solar System Science”. Johns Hopkins Earth and Planetary Sciences Department, Bromery Seminar, MD, Sept. 14
5. **Lustig-Yaeger, J.** (2022). “Rocky Exoplanet Characterization on the Path Towards Habitability and Biosignatures”. University of Maryland Planetary Astronomy Lunch Seminar Series, MD, Feb. 21
4. **Lustig-Yaeger, J.** & Meadows, V. S. (2019). “[The Era of Terrestrial Exoplanet Characterization](#)”. AGU, Joint AGU-AAS Session on Frontiers in Exoplanets, San Francisco, CA, 10 Dec. *Watch it on YouTube!*
3. **Lustig-Yaeger, J.**, Meadows, V. S., & Lincowski, A. P. (2019). “[Prospects for Biosignatures for Rocky Planets in the Era of JWST](#)”. SEEC Symposium 2019, “Rocky Exoplanets in the Era of JWST: Theory and Observation”, NASA Goddard, Greenbelt, MD, 5 Nov.
2. **Lustig-Yaeger, J.**, Meadows, V. S., & Lincowski, A. P. (2019). “[The Detectability and Characterization of the TRAPPIST-1 Exoplanet Atmospheres with JWST](#)”. ExoPAG 20, Bellevue, WA, 23 Jun.
1. **Lustig-Yaeger, J.**, Meadows, V., Tovar, G., Schwieterman, E., & Fujii, Y. “Prospects for Mapping Terrestrial Exoplanets with LUVOIR”. Goddard Space Flight Center, October 25, 2017

Contributed Talks

13. Contributed talk at Exoplanet Atmospheres 2026 in Denver, CO on “Charting the Cosmic Shoreline”
12. Contributed plenary talk at the conference *Towards the Habitable Worlds Observatory: Visionary Science and Transformational Technology (HWO25)*, July 2025.
11. **Lustig-Yaeger, J.**, Egan, A., Mayorga, L., et al. (2025). “[Venus as a Transiting Exoplanet: Retrievals on an Observed Venus Transmission Spectrum](#)”, American Astronomical Society Meeting Abstracts, 246, #215.02.
10. **Lustig-Yaeger, J.**, Cañas, C., Louie, D., et al. (2025). “[Looking at Giant Exoplanets around M-dwarfs \(GEMS\) with JWST: Constraining the Atmosphere of TOI-5205 b Through Strong Stellar Contamination](#)”, American Astronomical Society Meeting Abstracts, 245, #149.04.
9. **Lustig-Yaeger, J.**, Stevenson, K. B., et al. (2024). “[Voyage to the Cosmic Shoreline: JWST Transmission Spectroscopy of Five Rocky Exoplanets](#)”. American Astronomical Society Meeting Abstracts, 243, #444.06
8. **Lustig-Yaeger, J.**, Meadows, V. S., Crisp, D., Line, M., & Robinson, T. (2022). “Earth as a Transiting Exoplanet: Validating Transmission Spectroscopy Retrievals for Upcoming Terrestrial Exoplanet Habitability and Biosignature Searches”. Astrobiology Science Conference, Atlanta, GA, 15 - 20 May 2022.
7. **Lustig-Yaeger, J.**, Meadows, V. S., & Lincowski, A. P. (2019). “[TRAPPIST-1 and Beyond: Strategies for Characterizing Terrestrial Exoplanets and their Habitability](#)”. Abstract [202-7] presented at 2019 Astrobiology Science Conference, Bellevue, WA, 24-28 June.
6. **Lustig-Yaeger, J.**, Meadows, V. S., & Lincowski, A. P. (2019). “Detecting and Characterizing Terrestrial Atmospheres in the TRAPPIST-1 System with JWST”. TRAPPIST-1 Conference, Liège, Belgium, 11-14 Jun.

5. **Lustig-Yaeger, J.**, Meadows, V. S., & Lincowski, A. P. (2019). “[Simulating the Detectability and Characterization of the TRAPPIST-1 Exoplanet Atmospheres with JWST](#)”. American Astronomical Society Meeting Abstracts, 233, #103.03
4. **Lustig-Yaeger, J.**, Meadows, V. S., Tovar, G., Schwieterman, E., W., Fujii, Y., Luger, R., & Robinson, T. D. (2018). “[Detecting Oceans on Exoplanets with Next-Generation Telescopes](#)”. 2018 AGU Fall Meeting, Washington, D.C., 10-14 Dec. P51B-09
3. **Lustig-Yaeger, J.**, Luger, R., & Agol, E. (2017). “[Probing the Orbital Dynamics and Atmospheric Properties of the TRAPPIST-1 Planets with JWST](#)”. Habitable Worlds 2017, #4100
2. **Lustig-Yaeger, J.**, Tovar, G., Fujii, Y., Schwieterman, E., & Meadows, V. (2017). “[Mapping Surfaces and Clouds on Terrestrial Exoplanets Observed with Next-Generation Coronagraph-Equipped Telescopes](#)”. Astrobiology Science Conference, #3558
1. **Lustig-Yaeger, J.**, Line, M. R., & Fortney, J. J. (2015). “[On the Confidence of Molecular Detections in the Atmospheres of Exoplanets from Secondary Eclipse Spectra](#)”. American Astronomical Society Meeting Abstracts, 225, #124.03

Posters

10. Contributed poster presentation at AbSciCon 2026, Madison, WI.
9. **Lustig-Yaeger, J.**, Stevenson, K., Fu, G., May, E., Ceballos, K. O., Moran, S., et al. (2023). “[Constraints on the Presence of a Rocky Exoplanet Atmosphere from a JWST Transmission Spectrum](#)”. American Astronomical Society Meeting #241, id. #159.07.
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1. **Lustig-Yaeger, J.**, Fortney, J. J., Freedman, R., Marley, M. S., & Lupu, R. E. (2014). “[Gaseous Mean Opacities for Giant Planet and Brown Dwarf Atmospheres](#)”. American Astronomical Society Meeting Abstracts #223, #347.04

Public Talks

- “The Fate of Rocky Exoplanet Atmospheres and the Cosmic Shoreline” Astronomy on Tap, Madison, WI. May 20, 2026.
- “How Astronomers are Searching for Life on Planets Outside of the Solar System” Western Institute for Lifelong Learning, Silver City, NM. November 10, 2022.
- “Proxima Centauri b: A World of Possibilities” and panel discussion with Guillem Anglada-Escude, Rory Barnes, & Olivier Guyon, UW Astrobiology & the NASA Astrobiology Institute Lecture Series, Seattle, WA. May 3, 2017.

- “BREAKING: Terrestrial Exoplanet Discovered in the Habitable Zone of Proxima Centauri” Astronomy on Tap, Peddler Brewing Company, Seattle, WA. August 24, 2016.

Press

- Cutts, E. (2025, March 12). “The road map to alien life passes through the ‘Cosmic Shoreline’.” *Quanta Magazine*.
- Johns Hopkins Applied Physics Lab, Laurel, MD. (2023, January 11). “[NASA’s Webb Identifies Its First Exoplanet And It’s the Size of Earth](#)” [Press release].
 - AAS 241 Press Conference, [slides available here](#)
 - See also press releases and articles from: [NASA](#), [ESA](#), [Sky & Telescope](#)
- University of Washington, Seattle. (2019, August 13). “[James Webb Space Telescope could begin learning about TRAPPIST-1 atmospheres in a single year, study indicates](#)” [Press release].

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